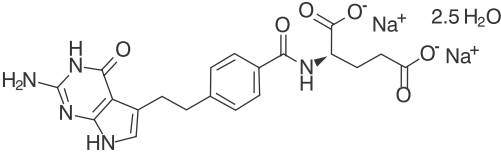


For the use of registered oncologists only.

RELITREXED 100
(Pemetrexed 100 mg powder for concentrate for solution for infusion)
RELITREXED 500
(Pemetrexed 500 mg powder for concentrate for solution for infusion)

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)
RELITREXED, is a white-to-light yellow or green-yellow lyophilized powder supplied in single-dose vials for reconstitution for intravenous infusion.
RELITREXED 100: Carton containing one (1) single-dose vial of 100 mg pemetrexed.
RELITREXED 500: Carton containing one (1) single-dose vial of 500 mg pemetrexed.

PRODUCT DESCRIPTION
RELITREXED is a folate analog metabolic inhibitor. The drug substance, pemetrexed disodium hemipentahydrate, has the chemical name L-glutamic acid, N-[4-[2-(2-amino-4,7-dihydro-4-oxo-1Hpyrrolo [2,3-d] pyrimidin-5yl) ethyl] benzoyl]-, Disodium, hemi-pentahydrate with a molecular formula of C₁₆H₁₆N₄Na₂O₇ 2.5 H₂O and a molecular weight of 516.37. The structural formula is as follows:



RELITREXED is a sterile white-to-light yellow or green-yellow lyophilized powder in single-dose vials to be reconstituted for intravenous infusion.

Before reconstitution:
White to either light yellow or green-yellow lyophilized powder or cake.

After reconstitution:
Clear and ranges in colour from colorless to yellow or green-yellow solution, practically free from visible particles.
(Diluents for reconstitution: Sodium chloride 9 mg/ml (0.9%) solution for injection)

Each 100-mg vial of **RELITREXED** contains pemetrexed disodium equivalent to 100 mg pemetrexed and 106 mg mannitol. Each 500 mg vial of **RELITREXED** contains pemetrexed disodium equivalent to 500 mg pemetrexed and 500 mg mannitol. Hydrochloric acid and/or sodium hydroxide may have been added to adjust pH.

PHARMACOLOGICAL PROPERTIES
Pharmacodynamic Properties
Pharmacotherapeutic group: Folic acid analogues, ATC code: L01BA04
RELITREXED is a multi-targeted anti-cancer antifolate agent that exerts its action by disrupting crucial folate-dependent metabolic processes essential for cell replication.
Pemetrexed behaves as a multitargeted antifolate by inhibiting thymidylate synthase (TS), dihydrofolate reductase (DHFR), and glycinamide ribonucleotide formyltransferase (GARFT), which are key folate-dependent enzymes for the de novo biosynthesis of thymidine and purine nucleotides. Pemetrexed is transported into cells by both the reduced folate carrier and membrane folate binding protein transport systems. Once in the cell, pemetrexed is rapidly and efficiently converted to polyglutamate forms by the enzyme folylpolyglutamate synthetase. The polyglutamate forms are retained in cells and are even more potent inhibitors of TS and GARFT. Polyglutamation is a timeand concentration-dependent process that occurs in tumour cells and, to a lesser extent, in normal tissues. Polyglutamated metabolites have an increased intracellular half-life resulting in prolonged drug action in malignant cells.

Pharmacokinetic Properties
Pemetrexed has a steady-state volume of distribution of 9 l/m2. Pemetrexed is approximately 81 % bound to plasma proteins. Binding was not notably affected by varying degrees of renal impairment. Pemetrexed undergoes limited hepatic metabolism. Pemetrexed is primarily eliminated in the urine, with 70 % to 90 % of the administered dose being recovered unchanged in urine within the first 24 hours following administration. Pemetrexed is actively secreted by OAT3 (organic anion transporter). Pemetrexed total systemic clearance is 91.8 ml/min and the elimination half-life from plasma is 3.5 hours in patients with normal renal function (creatinine clearance of 90 ml/min). Between patient variability in clearance is moderate at 19.3 %. Pemetrexed total systemic exposure (AUC) and maximum plasma concentration increase proportionally with dose. The pharmacokinetics of pemetrexed are consistent over multiple treatment cycles.

The pharmacokinetic properties of pemetrexed are not influenced by concurrently administered cisplatin. Oral folic acid and intramuscular vitamin B12 supplementation do not affect the pharmacokinetics of pemetrexed.

INDICATIONS
Malignant pleural mesothelioma
RELITREXED in combination with cisplatin is indicated for the treatment of chemotherapy naïve patients with unresectable malignant pleural mesothelioma.

Non-small cell lung cancer
RELITREXED in combination with cisplatin is indicated for the first line treatment of patients with locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology.

RELITREXED is indicated as monotherapy for the maintenance treatment of locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology in patients whose disease has not progressed immediately following platinum-based chemotherapy.

RELITREXED is indicated as monotherapy for the second line treatment of patients with locally advanced or metastatic non-small cell lung cancer other than predominantly squamous cell histology.

RELITREXED in combination with pembrolizumab and platinum chemotherapy, is indicated for the first line treatment of patients with metastatic non-squamous NSCLC, with no EGFR or ALK genomic tumor aberrations.

DOSAGE AND ADMINISTRATION
Posology
RELITREXED must only be administered under the supervision of a physician qualified in the use of anti-cancer chemotherapy.
RELITREXED in combination with cisplatin
The recommended dose of **RELITREXED** is 500 mg/m² of body surface area (BSA) administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle. The recommended dose of cisplatin is 75 mg/m² BSA infused over two hours approximately 30 minutes after completion of the pemetrexed infusion on the first day of each 21-day cycle. Patients must receive adequate antiemetic treatment and appropriate hydration prior to and/or after receiving cisplatin (see also cisplatin product insert for specific dosing advice).

RELITREXED as single agent
In patients treated for non-small cell lung cancer after prior chemotherapy, the recommended dose of **RELITREXED** is 500 mg/m² BSA administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle.

RELITREXED in combination with pembrolizumab
The recommended dose of **RELITREXED** when administered with pembrolizumab and platinum chemotherapy for the initial treatment of metastatic non-squamous NSCLC in patients with a creatinine clearance (calculated by Cockcroft-Gault equation) of 45 mL/min or greater is 500 mg/m² as an intravenous infusion over 10 minutes administered after pembrolizumab and prior to carboplatin or cisplatin on Day 1 of each 21-day cycle for 4 cycles. Following completion of platinum-based therapy, treatment with **RELITREXED** with or without pembrolizumab is administered until disease progression or unacceptable toxicity (see also pembrolizumab product insert for specific dosing advice).

Pre-medication regimen
To reduce the incidence and severity of skin reactions, a corticosteroid should be given the day prior to, on the day of, and the day after pemetrexed administration. The corticosteroid should be equivalent to 4 mg of dexamethasone administered orally twice a day.

To reduce toxicity, patients treated with pemetrexed must also receive vitamin supplementation. Patients must take oral folic acid or a multivitamin containing folic acid (350 to 1000 micrograms) on a daily basis. At least five doses of folic acid must be taken during the seven days preceding the first dose of pemetrexed, and dosing must continue during the full course of therapy and for 21 days after the last dose of pemetrexed. Patients must also receive an intramuscular injection of vitamin B12 (1000 micrograms) in the week preceding the first dose of pemetrexed and once every three cycles thereafter. Subsequent vitamin B12 injections may be given on the same day as pemetrexed.

Monitoring
Patients receiving pemetrexed should be monitored before each dose with a complete blood count, including a differential white cell count (WCC) and platelet count. Prior to each chemotherapy administration, blood chemistry tests should be collected to evaluate renal and hepatic function. Before the start of any cycle of chemotherapy, patients are required to have the following: absolute neutrophil count (ANC) should be ≥ 1500 cells/mm³ and platelets should be ≥ 100,000 cells/mm³. Creatinine clearance should be ≥ 45 ml/min. The total bilirubin should be ≤ 1.5 times upper limit of normal. Alkaline phosphatase (AP), aspartate aminotransferase (AST or SGOT) and alanine aminotransferase (ALT or SGPT) should be ≤ 3 times upper limit of normal. Alkaline phosphatase, AST and ALT ≤ 5 times upper limit of normal is acceptable if liver has tumour involvement.

Dose adjustments
Dose adjustments at the start of a subsequent cycle should be based on nadir haematologic counts or maximum non-haematologic toxicity from the preceding cycle of therapy. Treatment may be delayed to allow sufficient time for recovery. Upon recovery patients should be retreated using the guidelines in Tables 1, 2 and 3, which are applicable for **RELITREXED** used as a single agent or in combination with cisplatin.

Table 1 - Dose modification table for RELITREXED (as single agent or in combination) and cisplatin-Haematologic toxicities	
Nadir ANC < 500/mm ³ and nadir platelets ≥ 50,000/mm ³	75 % of previous dose (both RELITREXED and cisplatin)
Nadir platelets < 50,000/mm ³ regardless of nadir ANC	75 % of previous dose (both RELITREXED and cisplatin)
Nadir platelets < 50,000/mm ³ with bleeding ^a , regardless of nadir ANC	50 % of previous dose (both RELITREXED and cisplatin)

^a These criteria meet the National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998) definition of ≥ CTC Grade 2 bleeding
If patients develop non-haematologic toxicities ≥ Grade 3 (excluding neurotoxicity), **RELITREXED** should be withheld until resolution to less than or equal to the patient's pre-therapy value. Treatment should be resumed according to the guidelines in Table 2.

Table 2 - Dose modification table for RELITREXED (as single agent or in combination) and cisplatin-Non-haematologic toxicities ^{a,b}		
	Dose of RELITREXED (mg/m ²)	Dose of cisplatin (mg/m ²)
Any Grade 3 or 4 toxicities except mucositis	75 % of previous dose	75 % of previous dose
Any diarrhoea requiring hospitalisation (irrespective of grade) or grade 3 or 4 diarrhoea	75 % of previous dose	75 % of previous dose
Grade 3 or 4 mucositis	50 % of previous dose	100 % of previous dose

^a National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998)^b Excluding neurotoxicity
In the event of neurotoxicity, the recommended dose adjustment for **RELITREXED** and cisplatin is documented in Table 3. Patients should discontinue therapy if Grade 3 or 4 neurotoxicity is observed.

Table 3 - Dose modification table for RELITREXED (as single agent or in combination) and cisplatin-Neurotoxicity		
CTC ^a Grade	Dose of RELITREXED (mg/m ²)	Dose of cisplatin (mg/m ²)
0 - 1	100 % of previous dose	100 % of previous dose
2	100 % of previous dose	50 % of previous dose

^a National Cancer Institute Common Toxicity Criteria (CTC v2.0; NCI 1998)
Treatment with **RELITREXED** should be discontinued if a patient experiences any haematologic or non-haematologic Grade 3 or 4 toxicity after 2 dose reductions or immediately if Grade 3 or 4 neurotoxicity is observed.

Elderly
There has been no indication that patients 65 years of age or older are at increased risk of adverse events compared to patients younger than 65 years old. No dose reductions other than those recommended for all patients are necessary.

Paediatric population
There is no relevant use of **RELITREXED** in the paediatric population in malignant pleural mesothelioma and non-small cell lung cancer.

Patients with renal impairment (Standard Cockcroft and Gault formula or Glomerular Filtration Rate measured Tc99m-DPTA serum clearance method) Pemetrexed is primarily eliminated unchanged by renal excretion. Patients with creatinine clearance of ≥ 45 ml/min required no dose adjustments other than those recommended for all patients. There are insufficient data on the use of pemetrexed in patients with creatinine clearance below 45 ml/min; therefore the use of pemetrexed is not recommended.

Patients with hepatic impairment
No relationships between AST (SGOT), ALT (SGPT), or total bilirubin and pemetrexed pharmacokinetics were identified. However patients with hepatic impairment such as bilirubin > 1.5 times the upper limit of normal and/or aminotransferase > 3.0 times the upper limit of normal (hepatic metastases absent) or > 5.0 times the upper limit of normal (hepatic metastases present) have not been specifically studied.
Method of administration
For Precautions to be taken before handling or administering **RELITREXED**, see Special precautions for disposal and other handling.
RELITREXED should be administered as an intravenous infusion over 10 minutes on the first day of each 21-day cycle. For instructions on reconstitution and dilution of **RELITREXED** before administration, see Special precautions for disposal and other handling.

CONTRAINDICATIONS
Hypersensitivity to the active substance or to any of the excipients.

Breast-feeding
It is not known whether pemetrexed is excreted in human milk and adverse reactions on the suckling child cannot be excluded. Breast-feeding must be discontinued during pemetrexed therapy.

Concomitant use contraindicated: Yellow fever vaccine: risk of fatal generalised vaccinal disease.
WARNINGS AND PRECAUTIONS
Pemetrexed can suppress bone marrow function as manifested by neutropenia, thrombocytopenia and anaemia (or pancytopenia). Myelosuppression is usually the dose-limiting toxicity. Patients should be monitored for myelosuppression during therapy and pemetrexed should not be given to patients until absolute neutrophil count (ANC) returns to ≥ 1500 cells/mm³ and platelet count returns to ≥ 100,000 cells/mm3. Dose reductions for subsequent cycles are based on nadir ANC, platelet count and maximum non-haematologic toxicity seen from the previous cycle.

Less toxicity and reduction in Grade 3/4 haematologic and non-haematologic toxicities such as neutropenia, febrile neutropenia and infection with Grade 3/4 neutropenia were reported when pretreatment with folic acid and vitamin B12 was administered. Therefore, all patients treated with pemetrexed must be instructed to take folic acid and vitamin B12 as a prophylactic measure to reduce treatment-related toxicity.

Skin reactions have been reported in patients not pre-treated with a corticosteroid. Pre-treatment with dexamethasone (or equivalent) can reduce the incidence and severity of skin reactions.

The use of pemetrexed in patients with creatinine clearance of < 45 ml/min is not recommended.

Patients with mild to moderate renal insufficiency (creatinine clearance from 45 to 79 ml/min) should avoid taking non-steroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen, and acetylsalicylic acid (> 1.3 g daily) for 2 days before, on the day of, and 2 days following pemetrexed administration.
In patients with mild to moderate renal insufficiency eligible for pemetrexed therapy NSAIDs with long elimination half-lives should be interrupted for at least 5 days prior to, on the day of, and at least 2 days following pemetrexed administration.
Serious renal events, including acute renal failure, have been reported with pemetrexed alone or in association with other chemotherapeutic agents. Many of the patients in whom these occurred had underlying risk factors for the development of renal events including dehydration or pre-existing hypertension or diabetes. Nephrogenic diabetes insipidus and renal tubular necrosis were also reported in post marketing setting with pemetrexed alone or with other chemotherapeutic agents.
Most of these events resolved after pemetrexed withdrawal. Patients should be regularly monitored for acute tubular necrosis, decreased renal function and signs and symptoms of nephrogenic diabetes insipidus (e.g. hypernatraemia).
The effect of third space fluid, such as pleural effusion or ascites, on pemetrexed is not fully defined. Thus, drainage of third space fluid collection prior to pemetrexed treatment should be considered, but may not be necessary.
Due to the gastrointestinal toxicity of pemetrexed given in combination with cisplatin, severe dehydration has been observed. Therefore, patients should receive adequate antiemetic treatment and appropriate hydration prior to and/or after receiving treatment.
Serious cardiovascular events, including myocardial infarction and cerebrovascular events have been uncommonly reported with pemetrexed, usually when given in combination with another cytotoxic agent. Most of the patients in whom these events have been observed had pre-existing cardiovascular risk factors.
Immunodepressed status is common in cancer patients. As a result, concomitant use of live attenuated vaccines is not recommended.
Pemetrexed can have genetically damaging effects. Sexually mature males are advised not to father a child during the treatment and up to 6 months thereafter. Contraceptive measures or abstinence are recommended. Owing to the possibility of pemetrexed treatment causing irreversible infertility, men are advised to seek counselling on sperm storage before starting treatment.
Women of childbearing potential must use effective contraception during treatment with pemetrexed.
Cases of radiation pneumonitis have been reported in patients treated with radiation either prior, during or subsequent to their pemetrexed therapy. Particular attention should be paid to these patients and caution exercised with use of other radiosensitising agents.
Cases of radiation recall have been reported in patients who received radiotherapy weeks or years previously.
Excipients:
RELITREXED 100 (Pemetrexed 100 mg powder for concentrate for solution for infusion) This medicinal product contains less than 1 mmol sodium (23 mg) per vial, i.e. essentially 'sodiumfree'.
RELITREXED 500 (Pemetrexed 500 mg powder for concentrate for solution for infusion) This medicinal product contains 54 mg sodium per vial, equivalent to 2.7 % of the WHO recommended maximum daily intake of 2 g sodium for an adult.
To be taken into consideration by patients on a controlled sodium diet.

ADVERSE REACTIONS
Summary of the safety profile
The most commonly reported undesirable effects related to pemetrexed, whether used as monotherapy or in combination, are bone marrow suppression manifested as anaemia, neutropenia, leukopenia, thrombocytopenia; and gastrointestinal toxicities, manifested as anorexia, nausea, vomiting, diarrhoea, constipation, pharyngitis, mucositis, and stomatitis. Other undesirable effects include renal toxicities, increased aminotransferases, alopecia, fatigue, dehydration, rash, infection/sepsis and neuropathy. Rarely seen events include Stevens-Johnson syndrome and Toxic epidermal necrolysis.

Tabulated list of adverse reactions
The table below provides the frequency grouping and severity of undesirable effects that have been reported in patients with Mesothelioma who receive pemetrexed in combination with cisplatin. These chemo-naïve patients were fully supplemented with folic acid and vitamin B₁₂. Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

System organ class	Frequency	Event*
Blood and lymphatic system disorders	Very Common	Neutrophils / Granulocytes decreased
		Leucocytes decreased
		Hemoglobin decreased
		Platelets decreased
Metabolism and nutrition disorders	Common	Dehydration
Nervous system disorders	Very Common	Neuropathy - Sensory
	Common	Taste disturbance
Eye Disorder	Common	Conjunctivitis
Gastrointestinal disorders	Very Common	Diarrhoea
		Vomiting
		Stomatitis / Pharyngitis
		Nausea
	Common	Anorexia
		Constipation
Skin and subcutaneous tissue disorders	Very Common	Dyspepsia
		Rash
		Alopecia
Renal and urinary disorders	Very Common	Creatinine elevation
General disorders and administration site conditions	Very Common	Creatinine clearance decreased**
		Fatigue

*Refer to National Cancer Institute CTC version 2 for each grade of toxicity except the term "creatinine clearance decreased"
** which is derived from the term "renal/genitourinary other".
Clinically relevant CTC toxicities were reported in patients receiving cisplatin and pemetrexed include: renal failure, infection, pyrexia, febrile neutropenia, increased AST, ALT, and GGT, urticaria, chest pain, arrhythmia and motor neuropathy.
The table below provides the frequency grouping and severity of undesirable effects that have been reported in patients with locally advanced or metastatic non-small cell lung cancer and received prior chemotherapy who received single agent Pemetrexed with folic acid and vitamin B12 supplementation.

System organ class	Frequency	Event*
Blood and lymphatic system disorders	Very Common	Neutrophils / Granulocytes decreased
		Leucocytes decreased
	Common	Hemoglobin decreased Platelets decreased
Gastrointestinal disorders	Very Common	Diarrhoea Vomiting Stomatitis / Pharyngitis Nausea Anorexia
		Constipation
	Common	SGPT (ALT) elevation SGOT (AST) elevation
Hepatobiliary disorders	Common	Rash / desquamation
	Very Common	Pruritus Alopecia
Skin and sub- cutaneous tissue disorders	Very Common	Fatigue
	Common	Fever
General disorders and administration site conditions	Very Common	

*Refer to National Cancer Institute CTC version 2 for each grade of toxicity.
Clinically relevant CTC toxicities were reported in patients receiving pemetrexed include: infection without neutropenia, febrile neutropenia, allergic reaction/hypersensitivity, increased creatinine, motor neuropathy, sensory neuropathy, erythema multiforme, abdominal pain, and supraventricular arrhythmias.
The table below provides the frequency grouping and severity of undesirable effects that have been reported in patients with NSCLC who receive combination of cisplatin with pemetrexed. as initial treatment for locally advanced or metastatic NSCLC and fully supplemented with folic acid and vitamin B12.

System organ class	Frequency	Event*
Blood and lymphatic system disorders	Very Common	Hemoglobin decreased Neutrophils / Granulocytes decreased Leukocytes decreased
		Platelets decreased
		Neuropathy - sensory Taste disturbance
Nervous system disorders	Common	Nausea Vomiting Anorexia Constipation Stomatitis / Pharyngitis Diarrhoea without colostomy
		Dyspepsia / Heartburn
	Very Common	Alopecia
Skin and sub- cutaneous tissue disorders	Common	Rash/desquamation
	Very Common	Creatinine elevation
General disorders and administration site conditions	Very Common	Fatigue
	Common	

**Refer to National Cancer Institute CTC (v2.0; NCI 1998) for each Grade of Toxicity.
Clinically relevant CTC toxicities were reported in patients receiving combination of cisplatin with pemetrexed include: AST increase, ALT increase, infection, febrile neutropenia, renal failure, pyrexia, dehydration, conjunctivitis, creatinine clearance decrease, GGT increase, chest pain, arrhythmia, and motor neuropathy.
Clinically relevant toxicities with respect to gender were similar to the overall population in patients receiving pemetrexed plus cisplatin.
The table below provides the frequency grouping and severity of undesirable effects considered possibly related in patients receiving single-agent pemetrexed fully supplemented with folic acid and vitamin B12.
All patients were diagnosed with Stage IIIB or IV NSCLC and had received prior platinum based chemotherapy.

System organ class	Frequency	Event*
Blood and lymphatic system disorders	Very Common	Hemoglobin decreased
	Common	Leukocytes decreased Neutrophils decreased Neuropathy decreased
		Neuropathy - sensory
Nervous system disorders	Common	Nausea Anorexia Vomiting Mucositis / Stomatitis
		ALT (SGPT) elevation AST (SGOT) elevation
	Very Common	Rash / desquamation
Hepatobiliary disorders	Common	Fatigue
	Very Common	Pain Edema
Skin and sub- cutaneous tissue disorders	Common	Renal disorders****
	Common	

Abbreviations: ALT = alanine aminotransferase; AST = aspartate aminotransferase; CTCAE = Common Terminology Criteria for Adverse Event; NCI = National Cancer Institute; SGOT = serum glutamic oxaloacetic aminotransferase; SGPT = serum glutamic pyruvic aminotransferase.
** Refer to NCI CTCAE Criteria (Version 3.0; NCI 2003) for each grade of toxicity. The reporting rates shown are according to CTCAE version 3.0.
**** Combined term includes increased serum/blood creatinine, decreased glomerular filtration rate, renal failure and renal/genitourinary- other.
Clinically relevant CTC toxicity of any grade were reported in patients receiving pemetrexed include: febrile neutropenia, infection, decreased platelets, diarrhoea, constipation, alopecia, pruritus/itching, fever (in the absence of neutropenia), ocular surface disease (including conjunctivitis), increased lacrimation, dizziness, motor neuropathy, allergic reaction/hypersensitivity, erythema multiforme, supraventricular arrhythmia and pulmonary embolism. Serious cardiovascular and cerebrovascular events, including myocardial infarction, angina pectoris, cerebrovascular accident and transient ischaemic attack have been uncommonly reported, usually when given in combination with another cytotoxic agent. Most of the patients in whom these events have been observed had pre-existing cardiovascular risk factors.
Rare cases of hepatitis, potentially serious, have been reported.
Pancytopenia has been uncommonly reported.
Cases of colitis (including intestinal and rectal bleeding, sometimes fatal, intestinal perforation, intestinal necrosis and typhilitis) have been reported uncommonly in patients treated with pemetrexed.
Cases of interstitial pneumonitis with respiratory insufficiency, sometimes fatal, have been reported uncommonly in patients treated with pemetrexed.
Uncommon cases of oedema have been reported in patients treated with pemetrexed.
Oesophagitis/ radiation oesophagitis has been uncommonly reported.
Sepsis, sometimes fatal, has been commonly reported.
During post marketing surveillance, the following adverse reactions have been reported in patients treated with pemetrexed: Uncommon cases of acute renal failure have been reported with pemetrexed alone or in association with other chemotherapeutic agents Nephrogenic diabetes insipidus and renal tubular necrosis have been reported in post marketing setting with an unknown frequency.
Uncommon cases of radiation pneumonitis have been reported in patients treated with radiation either prior, during or subsequent to their pemetrexed therapy.
Rare cases of radiation recall have been reported in patients who have received radiotherapy previously. Uncommon cases of peripheral ischaemia leading sometimes to extremity necrosis have been reported.
Rare cases of bullous conditions have been reported including Stevens-Johnson syndrome and Toxic epidermal necrolysis in some cases were fatal.
Rarely, immune-mediated haemolytic anaemia has been reported in patients treated with pemetrexed.
Rare cases of anaphylactic shock have been reported.
Erythematous oedema mainly of the lower limbs has been reported with an unknown frequency.

DRUG INTERACTIONS

Pemetrexed is mainly eliminated unchanged renally by tubular secretion and to a lesser extent by glomerular filtration. Concomitant administration of nephrotoxic drugs (e.g. aminoglycoside, loop diuretics, platinum compounds, cyclosporin) could potentially result in delayed clearance of pemetrexed. This combination should be used with caution. If necessary, creatinine clearance should be closely monitored.
Concomitant administration of substances that are also tubularly secreted (e.g. probenecid, penicillin) could potentially result in delayed clearance of pemetrexed. Caution should be made when these drugs are combined with pemetrexed. If necessary, creatinine clearance should be closely monitored.
In patients with normal renal function (creatinine clearance ≥ 80 ml/min), high doses of non-steroidal anti-inflammatory drugs (NSAIDs, such as ibuprofen > 1600 mg/day) and acetylsalicylic acid at higher dose (≥ 1.3 g daily) may decrease pemetrexed elimination and, consequently, increase the occurrence of pemetrexed adverse events. Therefore, caution should be made when administering higher doses of NSAIDs or acetylsalicylic acid, concurrently with pemetrexed to patients with normal function (creatinine clearance ≥ 80 ml/min).
In patients with mild to moderate renal insufficiency (creatinine clearance from 45 to 79 ml/min), the concomitant administration of pemetrexed with NSAIDs (e.g. ibuprofen) or acetylsalicylic acid at higher dose should be avoided for 2 days before, on the day of, and 2 days following pemetrexed administration.
In the absence of data regarding potential interaction with NSAIDs having longer half-lives such as piroxicam or rofecoxib, the concomitant administration with pemetrexed in patients with mild to moderate renal insufficiency should be interrupted for at least 5 days prior to, on the day of, and at least 2 days following pemetrexed administration. If concomitant administration of NSAIDs is necessary, patients should be monitored closely for toxicity, especially myelosuppression and gastrointestinal toxicity.
Pemetrexed undergoes limited hepatic metabolism, Pemetrexed would not be predicted to cause clinically significant inhibition of the metabolic clearance of drugs metabolised by CYP3A, CYP2D6, CYP2C9, and CYP1A2.
Interactions common to all cytotoxics
Due to the increased thrombotic risk in patients with cancer, the use of anticoagulation treatment is frequent. The high intra-individual variability of the coagulation status during diseases and the possibility of interaction between oral anticoagulants and anticancer chemotherapy require increased frequency of INR (International Normalised Ratio) monitoring, if it is decided to treat the patient with oral anticoagulants.
Concomitant use contraindicated: Yellow fever vaccine: risk of fatal generalised vaccinal disease. Concomitant use not recommended: Live attenuated vaccines (except yellow fever, for which concomitant use is contraindicated); risk of systemic, possibly fatal, disease. The risk is increased in subjects who are already immunosuppressed by their underlying disease.

Use an inactivated vaccine where it exists (poliomyelitis)

Fertility, pregnancy and lactation

Contraception in males and females

Women of childbearing potential must use effective contraception during treatment with pemetrexed. Pemetrexed can have genetically damaging effects. Sexually mature males are advised not to father a child during the treatment and up to 6 months thereafter. Contraceptive measures or abstinence are recommended.

Pregnancy

There are no data from the use of pemetrexed in pregnant women but pemetrexed, like other anti-metabolites, is suspected to cause serious birth defects when administered during pregnancy. Animal studies have shown reproductive toxicity. Pemetrexed should not be used during pregnancy unless clearly necessary, after a careful consideration of the needs of the mother and the risk for the fetus.

Breast-feeding

It is not known whether pemetrexed is excreted in human milk and adverse reactions on the suckling child cannot be excluded. Breast-feeding must be discontinued during pemetrexed therapy.

Fertility

Owing to the possibility of pemetrexed treatment causing irreversible infertility, men are advised to seek counselling on sperm storage before starting treatment.

Effects on ability to drive and use machine

No studies on the effects on the ability to drive and use machines have been performed. However, it has been reported that RELITREXED may cause fatigue. Therefore patients should be cautioned against driving or operating machines if this event occurs.

Incompatibilities

RELITREXED is physically incompatible with diluents containing calcium, including lactated Ringer's injection and Ringer's injection. In the absence of other compatibility studies this medicinal product must not be mixed with other medicinal products.

OVERDOSAGE

Reported symptoms of overdose include neutropenia, anaemia, thrombocytopenia, mucositis, sensory polyneuropathy and rash. Anticipated complications of overdose include bone marrow suppression as manifested by neutropenia, thrombocytopenia and anaemia. In addition, infection with or without fever, diarrhoea, and/or mucositis may be seen. In the event of suspected overdose, patients should be monitored with blood counts and should receive supportive therapy as necessary. The use of calcium folinate/folinic acid in the management of pemetrexed overdose should be considered.

DOSAGE FORMS AND STRENGTHS

For infusion: 100 mg or 500 mg pemetrexed as a white to light-yellow or green-yellow lyophilized powder in single-dose vials for reconstitution.

Shelf life

Unopened vial: Refer to the expiry date on the product labels.

Reconstituted and infusion solutions

When prepared as directed, reconstituted and infusion solutions of **RELITREXED** contain no antimicrobial preservatives. Chemical and physical in-use stability of reconstituted and infusion solutions of pemetrexed were demonstrated for 24 hours at refrigerated temperature. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would not be longer than 24 hours at 2°C to 8°C.

Special precautions for storage

- Unopened vial
Store below 30°C. This medicinal product does not require any special storage conditions.
RELITREXED is a cytotoxic drug. Follow applicable special handling and disposal procedures
Special precautions for disposal and other handling
1. Use aseptic technique during the reconstitution and further dilution of pemetrexed for intravenous infusion administration.
2. Calculate the dose and the number of **RELITREXED** vials needed. Each vial contains an excess of pemetrexed to facilitate delivery of label amount.
3. **RELITREXED 100 mg**
Reconstitute 100-mg vials with 4.2 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection, without preservative, resulting in a solution containing 25 mg/ml pemetrexed.
RELITREXED 500 mg
Reconstitute 500-mg vials with 20 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection, without preservative, resulting in a solution containing 25 mg/ml pemetrexed.
Gently swirl each vial until the powder is completely dissolved. The resulting solution is clear and ranges in colour from colourless to yellow or green-yellow without adversely affecting product quality.
The pH of the reconstituted solution is between 6.6 and 7.8. **Further dilution is required.**
4. The appropriate volume of reconstituted pemetrexed solution must be further diluted to 100 ml with sodium chloride 9 mg/ml (0.9 %) solution for injection, without preservative, and administered as an intravenous infusion over 10 minutes.
5. Pemetrexed infusion solutions prepared as directed above are compatible with polyvinyl chloride and polyolefin lined administration sets and infusion bags.
6. Parenteral medicinal products must be inspected visually for particulate matter and discolouration prior to administration. If particulate matter is observed, do not administer.
7. Pemetrexed solutions are for single use only. Any unused medicinal product or waste material must be disposed of in accordance with local requirements.

Preparation and administration precautions

As with other potentially toxic anticancer agents, care should be exercised in the handling and preparation of pemetrexed infusion solutions. The use of gloves is recommended. If a pemetrexed solution contacts the skin, wash the skin immediately and thoroughly with soap and water. If pemetrexed solutions contact the mucous membranes, flush thoroughly with water. Pemetrexed is not a vesicant. There is not a specific antidote for extravasation of pemetrexed. There have been few reported cases of pemetrexed extravasation, which were not assessed as serious by the investigator. Extravasation should be managed by local standard practice as with other non-vesicants.

Product Registration Holder:

Mansa Healthcare Sdn. Bhd.
B-05-3A, 3 Two Square, No 2, Jalan 19/1, 46300 Petaling Jaya, Selangor, Malaysia.

Name and address of manufacturer:

Reliance Life Sciences Pvt Ltd.,
Plant 6, Dhirubhai Ambani Life Sciences Centre (DALC),
R-282 TTC area of MIDC, Thane – Belapur Road, Rabale, Navi Mumbai -400701, India

Date of revision of PI text

18 October 2023